Setting Up Your Computer

Learning Objectives

- Install some of the core software we will need in this class.
- Start becoming familiar with Command Line Interfaces (CLI)

Class Goals today:

A. Install Windows Subsystem Linux (WSL) on Windows machines
B. Explore Ubuntu Linux terminal and bash commands
C. Install miniconda package management system
D. Conda Commands
E. Create a virtual (Conda) environment from the OLCC2025.yml file
F. Connecting Conda Environments to Jupyter Lab and Jupyter Notebook
G. Install and configure VS Code

Note: If you have a Mac you do not need to install WSL/Ubuntu because MacOS Terminal uses a Unix-based shell (bash or zsh), which is very similar to Linux.

Note: These setup directions were from resources available through August 4, 2025. As technology is constantly changing, we may need to update these directions as we set up the computers. Please note that Dr. B. set his workstation up in June, and may not have the exact same setup as you.

Note: These directions are based on setup directions generated by Robert Belford at University of Arkansas, Little Rock as part of his Python for Science course. The original can be found here:

https://rebelford.github.io/IOCT-TECH/01_aSetUpComputer%20.html

A. Installing Linux via WSL

****NOTE: You do not Install WSL if you have a Mac****

Before installing WSL, ensure your computer meets these requirements:

Component	**Windows Users (for WSL2)**
OS Version	Windows 10 (version 1903+) or Windows 11
CPU	64-bit with virtualization support
RAM	At least 4 GB (8+ GB recommended) At least 8 GB RAM for basic usage and exercises in this class 16 GB RAM or more recommended for intensive machine learning activities
Disk Space	At least 10 GB free for tools & software

WSL will automatically install the Ubuntu distribution of Linux by default, but you can choose other distributions if desired. Note that while WSL can run on traditional HDDs, an SSD is recommended for better performance.

https://learn.microsoft.com/en-us/windows/wsl/install

https://learn.microsoft.com/en-us/windows/wsl/setup/environment

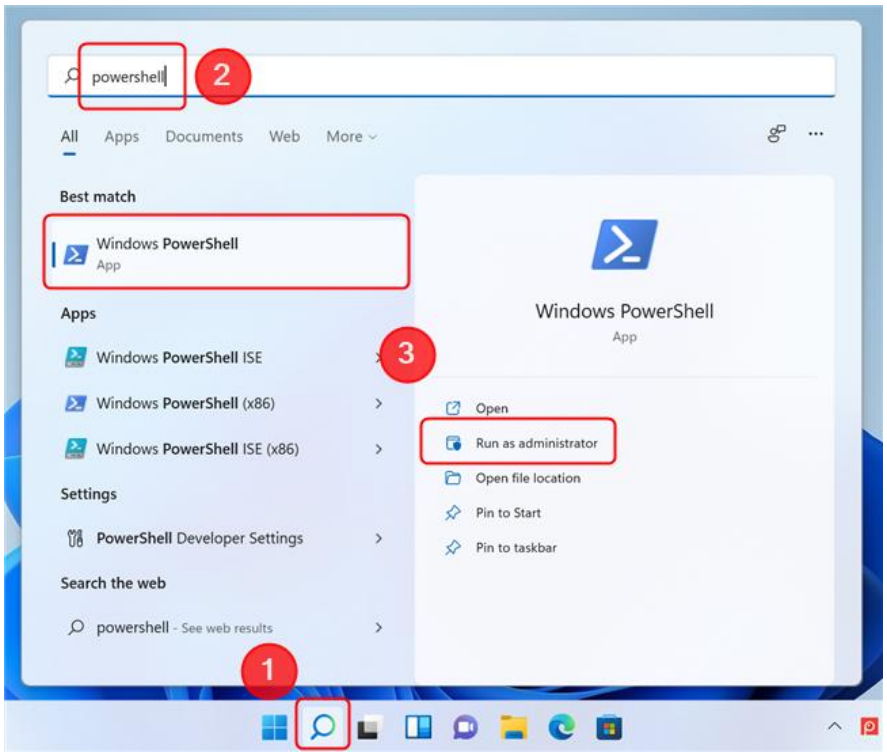
Why install WSL when you can use Windows?

Although python can be installed directly on Windows there are several reasons we are using WSL

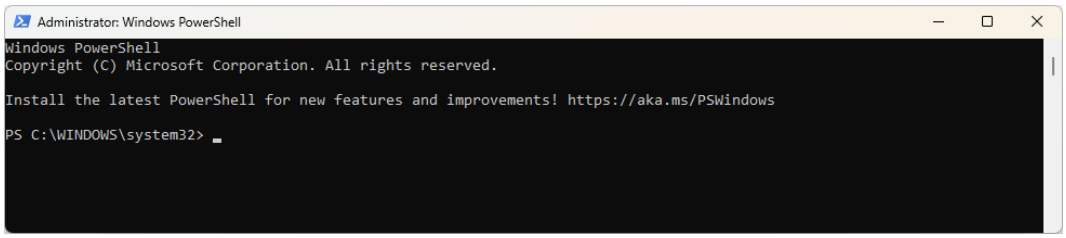
- Many Python Libraries are better supported with Linux OS than Windows
- Linux is a standard environment for open-source software production
- WSL allows Linux to integrate with Windows file system (in contrast to a dual boot setup).
- Python scripts often run faster on WSL compared to native Windows setup
- Learning Linux Bash scripts is a useful skill

1. Open Windows PowerShell or Windows Command Prompt in **administrator** mode by right-clicking and selecting "Run as administrator".

To open Windows PowerShell, use Search and type powershell. When you want to run it, click on the Windows PowerShell result. To run as administrator, you must right-click and select "Run as administrator".

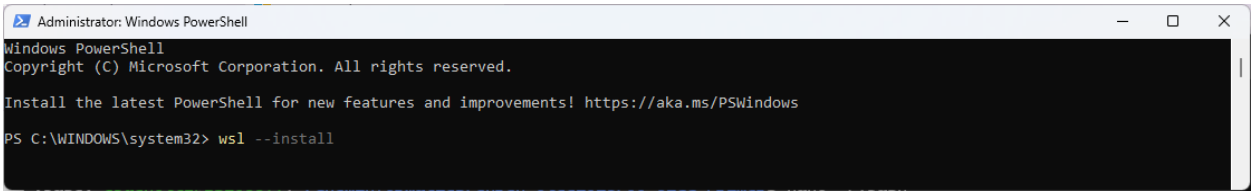


You should have a window that pops up that looks like this:



2. In the PowerShell type:
   wsl –install

   Note: When you see Highlighted Text these will be commands you need to paste into PowerShell and eventually the Ubuntu operating system.



This command will enable the features necessary to run WSL and install the Ubuntu distribution of Linux.

The `--install` command performs the following actions:

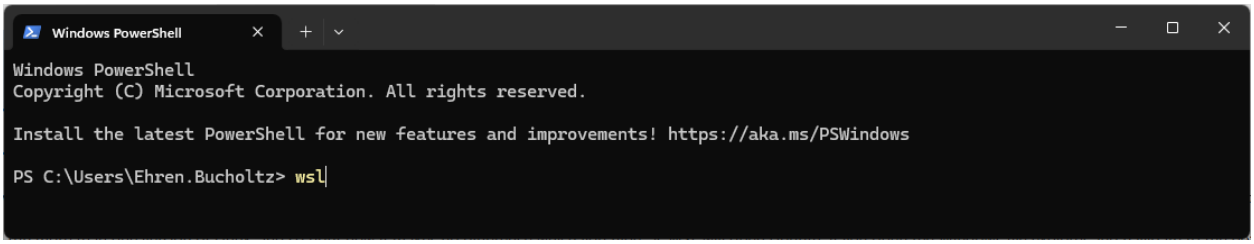
- Enables the optional WSL and Virtual Machine Platform components
- Downloads and installs the latest Linux kernel
- Sets WSL 2 as the default
- Downloads and installs the Ubuntu Linux distribution (reboot may be required)

You will need to restart your machine during this installation process.
Check the troubleshooting installation article if you run into any issues.

3. Once installed you can start Ubuntu through the Windows search bar (by typing Ubuntu).

Note, Typing either ubuntu or WSL will start Ubuntu on your windows machine as Ubuntu is the only version of Linux we are running. To avoid issues if another version is installed it is suggested that you always start with Ubuntu, and not WSL

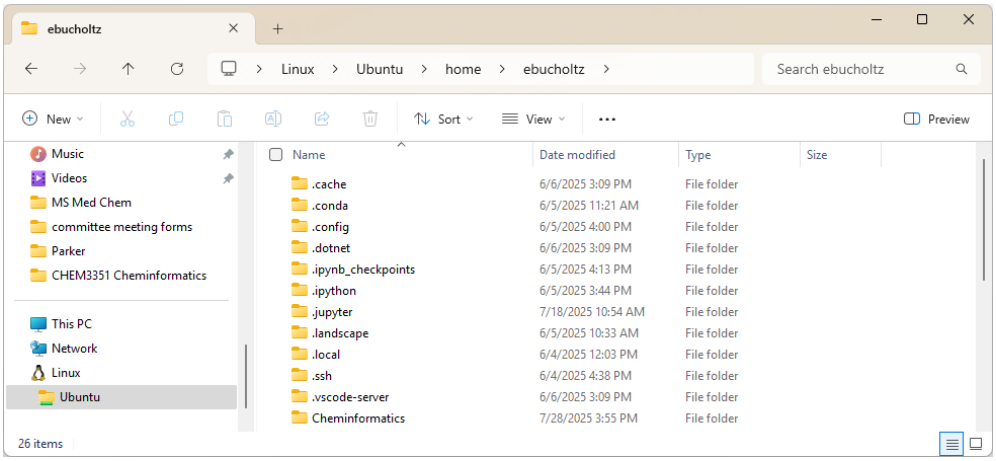
Alternatively, you can start Ubuntu through a Windows PowerShell as yourself (not administrator) and type **wsl.**



If you get an error like:

```
WslRegisterDistribution failed with error: 0x80370102 Please enable the
Virtual Machine Platform Windows feature and ensure virtualization is
enabled in the BIOS.
```

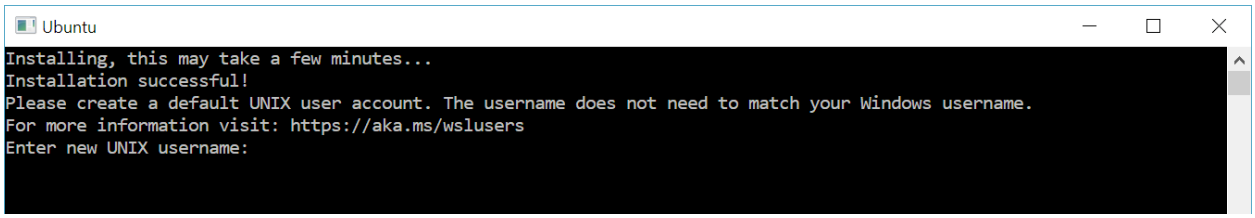
Click here if you get this error message.



WSL allows you to run more than one version of Linux, and so right now it makes no difference which you type, as WSL will open Ubuntu. If you open the Windows file explorer and navigate to the bottom you can see the directory structure and how WSL is at the very bottom. There will be a Linux Icon at the bottom of your file explorer and your files can be found in the path \\wsl.localhost\Ubuntu\home\username (e.g. ebucholtz). It is in this folder where you will be saving all your course materials.

4. Once you have installed WSL, you will need to create a user account and password for your newly installed Linux distribution.

In the search



Set up your Linux username and password

Once the process of installing your Linux distribution with WSL is complete, open the distribution (Ubuntu by default) using the Start menu. You will be asked to create a **User Name** and **Password** for your Linux distribution.

- This **User Name** and **Password** is specific to each separate Linux distribution that you install and has no bearing on your Windows username.
- Please note that while entering the **Password**, nothing will appear on screen. This is called *blind typing*. You won't see what you are typing, this is completely normal.
- Once you create a **Username** and **Password**, the account will be your default user for the distribution and automatically sign-in on launch.
- This account will be considered the Linux administrator, with the ability to run `sudo` (Super User Do) administrative commands.
- Each Linux distribution running on WSL has its own Linux user accounts and passwords. You will have to configure a Linux user account every time you add a distribution, reinstall, or reset.

If you need to change or reset your password, or you have forgotten it, click here to learn more.

B. Explore Ubuntu Linux terminal and bash commands

Before we install Miniconda3 we need to update our packages and versions. Note, we could use Anaconda or Anaconda Navigator, but in this class you need to use Miniconda. Miniconda is a minimal installation of Anaconda, which comes with many packages preinstalled and I want you to learn how to install and maintain your packages. Anaconda Navigator is a GUI (Graphical User Interface) for running Anaconda and I want you to gain experience with CLI (Command Line Interfaces), I also find it to be slower to load, and wastes computer resources.

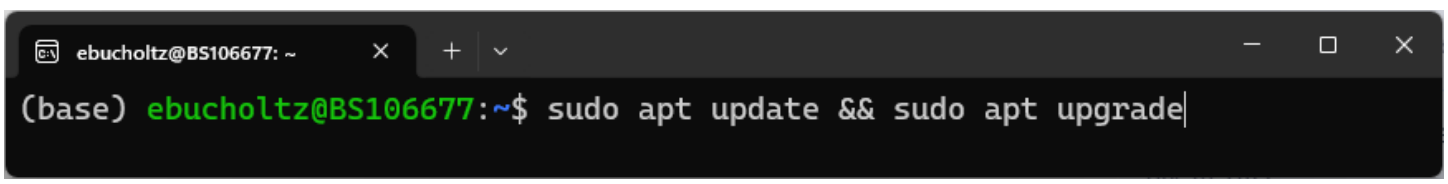
1. Update Packages

In your Ubuntu terminal, type sudo apt update && sudo apt upgrade

This command does two things:

- *sudo apt update*: Updates the list of available Linux packages and their versions. You may need your password the first time you run this
- *sudo apt upgrade*: Upgrades the installed packages to their latest versions.

Running this before installing new software ensures you're working with the most up-to-date linux system packages, which can prevent compatibility issues.



Note: my terminal says (base) because I have already installed miniconda.

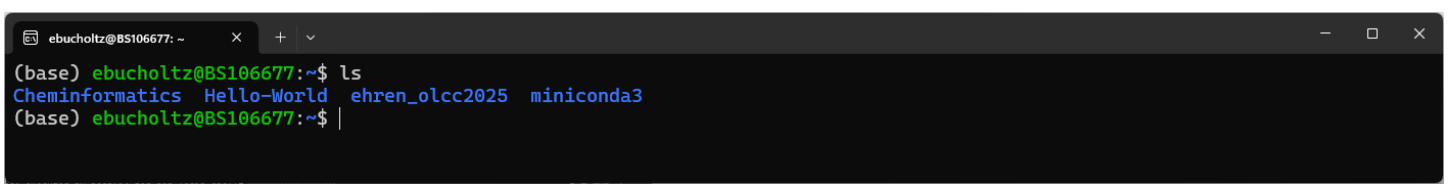
Explanation of the update command:

- *sudo*: Gives you temporary administrative privileges. (Super User Do
- *apt*: The package management tool for Ubuntu.
- *&&*: Runs the second command only if the first one succeeds.

2. Update Bash to always start in your Linux home directory and not your Windows directory

In your linux terminal, we are going to update the configuration script for the Bash (Bourne Again Shell). It is a hidden file called .bashrc that is located in a user's home directory.

To demonstrate it is hidden, first type ls in your terminal:



You can see the output here indicates that I have 4 directories (folder) as indicated by the color blue.

When I type `ls -al`, you can see that I have many hidden files.

```
ebucholtz@BS106677: ~
(base) ebucholtz@BS106677:~$ ls -al
total 132
drwxr-x--- 17 ebucholtz ebucholtz  4096 Jul 31 18:28 .
drwxr-xr-x  3 root      root       4096 Jun  4 11:53 ..
-rw-------  1 ebucholtz ebucholtz 18932 Jul 31 19:15 .bash_history
-rw-r--r--  1 ebucholtz ebucholtz   220 Jun  4 11:53 .bash_logout
-rw-r--r--  1 ebucholtz ebucholtz  4426 Jun  4 16:50 .bashrc
drwx------  6 ebucholtz ebucholtz  4096 Jun  6 15:09 .cache
drwxr-xr-x  2 ebucholtz ebucholtz  4096 Jun  5 11:21 .conda
drwxr-xr-x  4 ebucholtz ebucholtz  4096 Jun  5 16:00 .config
drwxr-xr-x  3 ebucholtz ebucholtz  4096 Jun  6 15:09 .dotnet
-rw-r--r--  1 ebucholtz ebucholtz    92 Jun  4 15:55 .gitconfig
drwxr-xr-x  2 ebucholtz ebucholtz  4096 Jun  5 16:13 .ipynb_checkpoints
drwxr-xr-x  3 ebucholtz ebucholtz  4096 Jun  5 15:44 .ipython
drwxr-xr-x  4 ebucholtz ebucholtz  4096 Jul 18 10:54 .jupyter
drwxr-xr-x  2 ebucholtz ebucholtz  4096 Jun  5 10:33 .landscape
-rw-------  1 ebucholtz ebucholtz    20 Jun  4 17:06 .lesshst
drwxr-xr-x  3 ebucholtz ebucholtz  4096 Jun  4 12:03 .local
-rw-rw-r--  1 ebucholtz ebucholtz     0 Jul 31 11:23 .motd_shown
-rw-r--r--  1 ebucholtz ebucholtz   807 Jun  4 11:53 .profile
drwx------  2 ebucholtz ebucholtz  4096 Jun  4 16:38 .ssh
-rw-r--r--  1 ebucholtz ebucholtz   317 Jun  4 12:11 .tcshrc
drwxr-xr-x  5 ebucholtz ebucholtz  4096 Jun  6 15:09 .vscode-server
-rw-r--r--  1 ebucholtz ebucholtz   268 Jul 31 11:26 .wget-hsts
-rw-r--r--  1 ebucholtz ebucholtz   683 Jun  4 12:11 .xonshrc
-rw-r--r--  1 ebucholtz ebucholtz   494 Jun  4 12:11 .zshrc
drwxr-xr-x  4 ebucholtz ebucholtz  4096 Jul 28 15:55 Cheminformatics
drwxr-xr-x  4 ebucholtz ebucholtz  4096 Jun  6 16:20 Hello-World
drwxr-xr-x  7 ebucholtz ebucholtz  4096 Jun  9 10:04 ehren_olcc2025
drwxr-xr-x 19 ebucholtz ebucholtz  4096 Jul 23 10:33 miniconda3
(base) ebucholtz@BS106677:~$
```

We are going to use a text editor to edit the .bashrc file

Type `nano ~/.bashrc` and press enter

```
ebucholtz@BS106677: ~
(base) ebucholtz@BS106677:~$ nano ~/.bashrc
```

To edit it, watch me demonstrate in class.

At the bottom of the file add `cd ~`

Save and close the file: Press Ctrl + O, then Enter, and finally Ctrl + X.

Next time you start WSL/Ubuntu it will start in your home directory.

Some Common File Management Bash Commands

You should use this section as a reference, and proceed to section (C) Installing Miniconda.

- pwd Print Working Directory (shows where you are in the file tree)
- ls list files and directories
  - ls -l -a long list including hidden files
  - ls ../ (reaches up one level)
- cd change directories
  - cd ~ navigate to home directory
  - cd / navigate to root directory
  - cd .. go up one level
  - cd - go down one level
- mkdir <directory_name> make a directory in current location
- rmdir <directory_name> remove a directory
  - rmdir -d <directory_name> remove empty directory
  - rmdir -r <directory_name> recursive, removes files and subfolders but asks for confirmation
  - rmdir -rf <directory_name> recursive and forced (does not ask for confirmation)
- rm <filename.extension removes files
- mv <file_name><directory_name> moves file to new directory
- exit closes terminal

NOTE: there are many bash commands associated with installing and managing packages and we will preferentially use Conda commands, like conda install <package-name>, although sometimes we will need to use bash commands like pip install <package-name> which involves the Python Package Index. Usually, you have both options and you should first try the conda option.

For the purposes of this course, I have tried to provide you with a file that will set up your environment and you should not need to do conda or pip installs.

C. Install miniconda package management system

First make sure you are in your user home directory (/home/username).

1. Type: pwd print working directory, which should return your user directory (where you are). If you are not in your user home directory (/home/username), type cd ~ to navigate to it
2. Type: ls - list files and subdirectories in current directory
3. Install Miniconda3 in your user directory. When you install miniconda here, a new directory will be created for miniconda. You do not need to make this directory yourself, it will automatically be created .

For Linux users:

Type: curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

Type: bash Miniconda3-latest-Linux-x86_64.sh

For Mac users: (note: I do not have a Mac, and so cannot test the instruction, but they seem logical to me)

Type: curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh

Type: bash Miniconda3-latest-MacOSX-x86_64.sh

- You will need to agree to the license and may have to scroll through it (use the down arrow).
- You will also need to confirm where location miniconda is being installed.
- You will also want to update the shell profile to initialize conda when the command prompt is activated.

Test your installation

1. Type: exit to close the terminal
2. Open Ubuntu terminal from the start menu
3. if you see (base) in front of the command prompt, conda is initialized.

If you do not see it, type conda init and you should now see (base) in front of the primary prompt.

conda command not found error: If you get this error, refer to article "Conda Command Not Found" by Peter Sawe. You will need to add the path to miniconda3 to your .bashrc file (bash run commands), and this will ensure that Conda and Python will use the versions installed by miniconda.

D. Common Conda Commands

- **Official Conda Cheat Sheet**
- **Official Conda Getting Started Tutorial** (Note: follow this tutorial below for creating your environment for this class).

Updating Conda

When you update conda in the base environment you want to run two commands in your terminal.

1. Type: conda update -n base -c defaults conda
2. Type: conda update --all

Now lets look at what each of these steps does.

conda update -n base -c defaults conda

This command updates conda itself in the base environment:

- conda update: This is the basic command to update packages.
- -n base: Specifies that the update should occur in the "base" environment, which is the default conda environment.
- -c defaults: Instructs conda to use the default channel for the update.
- conda: This is the package being updated, which is conda itself.

E. Create a virtual (Conda) environment from the OLCC2025.yml file

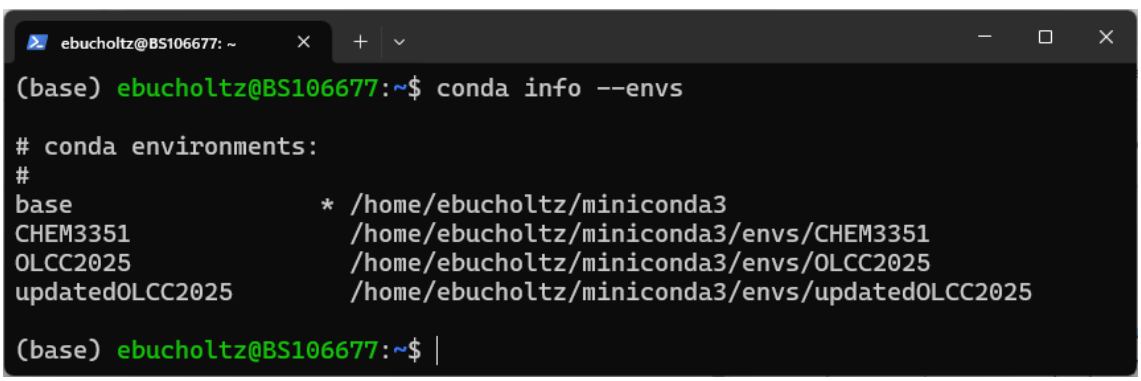
Overview of Virtual Environments and their Pieces

Our goal is to create a virtual environment we can access through a Jupyter Notebook and I will try and explain this at a high level before we get into creating our environment.

1. **The Environment** is like a container to hold your python interpreter and packages.
2. When you create an environment you create a **python interpreter**, and you can specify what version. You subsequently add the packages and their dependencies, and this is the environment you operate in. But you need to be able to access the environment.
3. In the context of Python, particularly with tools like Jupyter Notebooks, a **"kernel"** refers to the computational engine responsible for executing the code you write. It acts as an intermediary between the user interface (like the Jupyter Notebook or JupyterLab) and the Python interpreter. When you **register the kernel** you are giving Jupyter Lab the ability to find the kernel.

When you create a conda environment, conda will automatically create a unique folder for it in the miniconda3/envs folder. To see existing environments, run the following command.

```
conda info --envs
```



Note: The asterisk (*) in front of the base environment path indicates that it is the active environment. This is also indicated by the (base) in front of ebucholtz@BS106677:~$ in the command prompt. Also note that there are 3 other environments, and are in the envs folder.

Why have other environments?

The base environment houses Conda itself and its core dependencies. Installing project-specific packages directly into base increases the risk of dependency conflicts. Since different projects require different versions of the same package, it could result in potentially breaking the base environment or other projects.

Building an environment

Conda Channels
A Conda channel is a remote or local location where software packages are stored and from which the Conda package manager can retrieve and install them. Think of it as a repository or a directory containing various Conda packages.

Channel Priority
Before creating a channel we want to configure a few things in base environment that have global settings and will apply to all environments, unless an environment over rules them.

There are several channels that Anaconda uses to acquire channels and Conda Forge is considered as one of the best ones to give priority.

The following two lines of code give Conda Forge Priority. You should run these from the base environment.
Type: conda config --add channels conda-forge
Type: conda config --set channel_priority flexible

Creating an environment from a .yml file

A Conda environment .yml file, conventionally named environment.yml, is a YAML-formatted text file used to define and describe a Conda environment. (YAML originally stood for "Yet Another Markup Language" but was later changed to "YAML Ain't Markup Language") It is a human-readable data serialization format designed to be easily understandable by humans while also being readily parsed by machines. This file serves as a blueprint for creating, updating, or rebuilding a reproducible software environment.

Key components typically found in an environment.yml file:
- **name**: Specifies the name of the Conda environment.
- **channels**: Lists the Conda channels where packages should be searched for (e.g., defaults, conda-forge).
- **dependencies**: Lists the packages and their specific versions required for the environment. This can include packages installed via conda and potentially those installed via pip.

You have been supplied with a file called OLCC2025.yml. It is the blueprint for all the packages we need for the course this semester. We need to put this in a directory for the course in our linux distribution (Ubuntu.)

Type the following commands in your terminal.

cd	This will take you back to your home directory.
mkdir CHEM3351	This will create a directory called CHEM3351
cd CHEM3351	This will change the active directory to the new CHEM3351 directory
ls	This will give you the list of all files in the folder. (it should be empty)

Now to get the file type the following into your terminal:

wget https://raw.githubusercontent.com/ebucholtz/CHEM3351/refs/heads/main/CHEM3351.yml

> If you get an error, we need to add wget to retrieve content from web servers and and ca-certificates (to allow SSL-based applications to check for the authenticity of SSL connections), type:
>
> sudo apt-get install wget ca-certificates
>
> Then do the above wget statement again.

This will directly download the file to your active directory. To confirm type the following:

ls	This will give you the list of all files in the folder. It should have the .yml file.
nano CHEM3351.yml or	This will open the file and show you all the library dependencies are for the environment.

Now that we have the OLCC2025.yml file we need to make one final command to create our local environment.

Type: conda env create -f CHEM3351.yml -n CHEM3351

This will create an environment called CHEM3351based off the CHEM3351.yml file. For our class we are renaming it to our course code.

After creating your environment, type conda info --envs to list all your conda environments. You should see two environments: base and CHEM3351

Activating the environment

Type: conda activate CHEM3351. This command activates the environment. You should see a new prompt instead of base in front of your username.

Type: conda update --all

- conda update: Again, this is the basic update command.
- --all: This flag tells conda to update all packages in the current environment to their latest compatible versions.
- conda list --revisions gives the history of the current environment

The last thing we need to do is register the environment as a name so that we can easily recognize it as an ipython kernel.

Type: python -m ipykernel install --user --name=CHEM3351 --display-name "Python (CHEM3351)"

You should have something that looks like:

```
(CHEM3351) ebucholtz@BS106677:~$ python -m ipykernel install --user --name=CHEM3351 --display-name "Python (CHEM3351)"
Installed kernelspec CHEM3351 in /home/ebucholtz/.local/share/jupyter/kernels/chem3351
(CHEM3351) ebucholtz@BS106677:~$
```

The last thing we will do is just check the version of python we installed.

Type: python --version

The latest stable version of Python is 3.13.5, released on June 11, 2025. The Next major release will be 3.14, and a pre-release is expected on October 1, 2025.

F. Connecting Conda Environments to Jupyter Lab and Jupyter Notebook

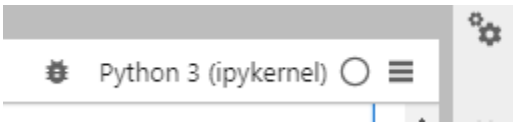
Now that you have a conda environment, we can launch jupyter lab or jupyter notebook to create workflows in that environment as those were installed with the other dependencies. JupyterLab and Jupyter Notebooks run as a local web application accessible through a web browser, typically on localhost. This means the server process is running on your own computer, and you access its interface using a web browser also on your local machine.

Type: jupyter lab

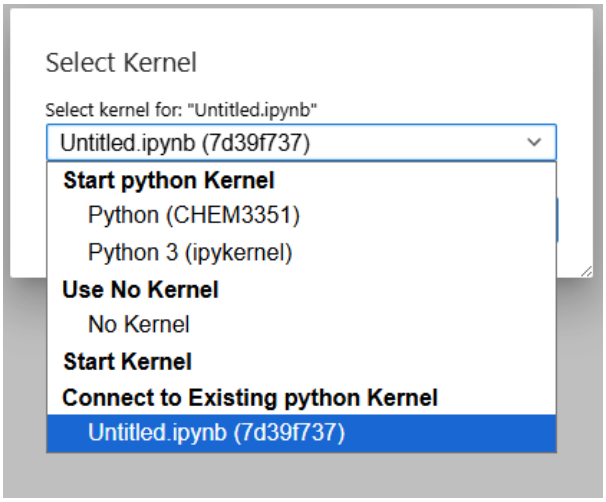
This should launch your default browser with Jupyter Lab running. If it doesn't, you can start your browser and in a new tab type localhost:8888/lab in the URL window. You may also have information in the terminal window that indicates a URL that copy/paste into the URL window.

To test that the system is working, in your browser, click File→New→Notebook

Now that you have Jupyter lab installed and have created your first environment, you can go to that environment and open the lab. But you need to connect the kernel to the one you registered in your environment, which is done by clicking on the box in the upper right corner of your notebook (Python3(ipkernel) in image below).



If you click it a drop down box will appear and you can choose the kernel you wish to use in this notebook. (I believe that Python 3 (ipykernel) is the same as CHEM3351 here, but I am not absolutely sure.)



To test that this kernel is working, we will do a test to make sure we can import a dependent library and see what version it is.

In your untitled notebook, copy and paste the following code into the first code cell:

```
import rdkit
```

```
print("The version of rdkit is:",rdkit.__version__)
```

You should have something returned like:

The version of rdkit is: 2025.03.5

Shut down your Jupyter notebook, and exit Linux for now.

G. Setting up the Visual Studio Code Editor

So far we have been using Jupyter notebooks to write and edit Python code. Not all coding projects are designed to run in a notebook environment. For those projects, we use a full-featured text editor. While we won't be running types of projects, we can take advantage of those text editors. We will be using Visual Studio Code (VS Code), which is a free editor developed by microsoft that supports not only python but also other languages and tools. It will allow us to continue to work with Jupyter notebooks. In addition, we will link it to our GitHub Copilot accounts we signed up for last week. This provides us with an AI-powered coding assistant that can suggest, generate and complete code, ultimately allows us to explore both AI-assisted and AI-generated coding as part of our cheminformatics workflow.

VS Code is available for all operating systems. Since this is a Microsoft product, and we are running Windows subsystem for linux, we are going to install it as a Windows application. We will follow the directions at https://learn.microsoft.com/en-us/windows/wsl/tutorials/wsl-vscode

1) Install VS Code and the WSL extension

- Visit the VS Code install page and select the 64 bit installer. Install Visual Studio Code on Windows (not in your WSL file system).
- When prompted to **Select Additional Tasks** during installation, be sure to check the **Add to PATH** option so you can easily open a folder in WSL using the code command.
- Install the Remote Development extension pack. This extension pack includes the WSL extension, in addition to the Remote - SSH, and Dev Containers extensions, enabling you to open any folder in a container, on a remote machine, or in WSL.

2) Update your Linux distribution

Some WSL Linux distributions are lacking libraries that are required by the VS Code server to start up. You can add additional libraries into your Linux distribution by using its package manager.

Open your Ubuntu distribution

Type: sudo apt-get update

To add wget (to retrieve content from web servers) and ca-certificates (to allow SSL-based applications to check for the authenticity of SSL connections), type:

sudo apt-get install wget ca-certificates

(Note we may have done this already)

To run VS Code in WSL:

Change to your directory you wish to code in. Many times you will set up folders for projects. I usually start my program from the directory I will be working.

Type: `code .` Yes, you need to add the period. It is essentially saying, run the program in the current working directory.

Extensions inside of VS Code WSL

The WSL extension splits VS Code into a “client-server” architecture, with the client (the user interface) running on your Windows machine and the server (your code, Git, plugins, etc) running "remotely" in your WSL distribution.

When running the WSL extension, selecting the 'Extensions' tab will display a list of extensions split between your local machine and your WSL distribution.

Installing a local extension, like a theme, only needs to be installed once.

Some extensions, like the Python extension or anything that handles things like linting or debugging, must be installed separately on each WSL distribution. VS Code will display a warning icon ⚠, along with a green "Install in WSL" button, if you have an extension locally installed that is not installed on your WSL distribution.

Click on the Extension Icon (note my icon looks like this as I chose the dark mode theme):

Let’s look at what is currently installed (both local-installed, and WSL:ubuntu installed)

We want to install the following:

- Python
- Jupyter
- WSL
- GitHub Copilot
- GitHub Copilot Chat

You can use the Search Extensions in the Marketplace to find them if they are not there.
Finally, we will have to link your github copilot credentials to the software.
https://code.visualstudio.com/docs/copilot/setup

Conclusion:

At this end of this setup, you should have installed WSL, set up a conda environment for the class, installed VS Code and are ready to use this system for local cheminformatics workflows on your computer.